

1. Let the tangent to the curve $x^2 + 2x - 4y + 9 = 0$ at the point $P(1, 3)$ on it meet the y -axis at A . Let the line passing through P and parallel to the line $x - 3y = 6$ meet the parabola $y^2 = 4x$ at B . If B lies on the line $2x - 3y = 8$, then $(AB)^2$ is equal to _____.

[2023 (06 Apr Shift 1)]

2. Let a curve $y = f(x)$, $x \in (0, \infty)$ pass through the points $P(1, \frac{3}{2})$ and $Q(a, \frac{1}{2})$. If the tangent at any point $R(b, f(b))$ to the given curve cuts the y -axis at the point $S(0, c)$ such that $bc = 3$, then $(PQ)^2$ is equal to _____.

[2023 (06 Apr Shift 2)]

3. The number of points, where the curve $y = x^5 - 20x^3 + 50x + 2$ crosses the x -axis, is _____.

[2023 (06 Apr Shift 2)]

4. A square piece of tin of side 30 cm is to be made into a box without top by cutting a square from each corner and folding up the flaps to form a box. If the volume of the box is maximum, then its surface area (in cm^2) is equal to

[2023 (10 Apr Shift 1)]

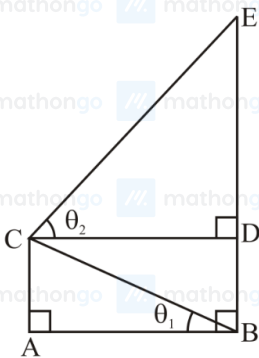
- (1) 800
(2) 675
(3) 1025
(4) 900

5. Let $g(x) = f(x) + f(1-x)$ and $f''(x) > 0$, $x \in (0, 1)$. If g is decreasing in the interval $(0, \alpha)$ and increasing in the interval $(\alpha, 1)$, then $\tan^{-1} 2\alpha + \tan^{-1}(\frac{1}{\alpha}) + \tan^{-1}(\frac{\alpha+1}{\alpha})$ is equal to

[2023 (10 Apr Shift 2)]

- (1) $\frac{\pi}{4}$
(2) $\frac{5\pi}{4}$
(3) $\frac{3\pi}{4}$
(4) $\frac{3\pi}{2}$

6. In the figure, $\theta_1 + \theta_2 = \frac{\pi}{2}$ and $\sqrt{3}(BE) = 4(AB)$. If the area of $\triangle CAB$ is $2\sqrt{3} - 3$ unit², when $\frac{\theta_2}{\theta_1}$ is the largest, then the perimeter (in unit) of $\triangle CED$ is equal to



[2023 (10 Apr Shift 2)]

7. Let the quadratic curve passing through the point $(-1, 0)$ and touching the line $y = x$ at $(1, 1)$ be $y = f(x)$. Then the x -intercept of the normal to the curve at the point $(\alpha, \alpha + 1)$ in the first quadrant is

[2023 (10 Apr Shift 2)]

8. Let $f : [2, 4] \rightarrow \mathbb{R}$ be a differentiable function such that $(x \log_e x)f'(x) + (\log_e x)f(x) + f(x) \geq 1$, $x \in [2, 4]$ with $f(2) = \frac{1}{2}$ and $f(4) = \frac{1}{2}$.

Consider the following two statements:

- (A) $f(x) \leq 1$, for all $x \in [2, 4]$
(B) $f(x) \geq 1/8$, for all $x \in [2, 4]$

Then,

[2023 (11 Apr Shift 1)]

- (1) Neither statement (A) nor statement (B) is true
(2) Only statement (B) is true
(3) Both the statements (A) and (B) are true
(4) Only statement (A) is true

9. If the total maximum value of the function $f(x) = \left(\frac{\sqrt{3}x}{2\sin x}\right)^{\sin^2 x}$, $x \in \left(0, \frac{\pi}{2}\right)$, is $\frac{k}{e}$, then $\left(\frac{k}{e}\right)^8 + \frac{k^8}{e^5} + k^8$ is equal to

[2023 (12 Apr Shift 1)]

- (1) $e^3 + e^6 + e^{11}$
- (2) $e^5 + e^6 + e^{11}$
- (3) $e^3 + e^6 + e^{10}$
- (4) $e^3 + e^5 + e^{11}$

10. $\max_{0 \leq x \leq \pi} \left\{ x - 2 \sin x \cos x + \frac{1}{3} \sin 3x \right\} =$

[2023 (13 Apr Shift 1)]

- (1) $\frac{\pi+2-3\sqrt{3}}{6}$
- (2) π
- (3) 0
- (4) $\frac{5\pi+2+3\sqrt{3}}{6}$

11. The set of all $a \in \mathbb{R}$ for which the equation $x|x-1| + |x+2| + a = 0$ has exactly one real root, is

[2023 (13 Apr Shift 1)]

- (1) $(-7, \infty)$
- (2) $(-\infty, \infty)$
- (3) $(-6, -3)$
- (4) $(-\infty, -3)$

12. Consider the triangles with vertices $A(2, 1)$, $B(0, 0)$ and $C(t, 4)$, $t \in [0, 4]$. If the maximum and the minimum perimeters of such triangles are obtained at $t = \alpha$ and $t = \beta$ respectively, then $6\alpha + 21\beta$ is equal to _____.

[2023 (15 Apr Shift 1)]

ANSWER KEYS

1. (292) 2. (5) 3. (5) 4. (1) 5. (1) 6. (6) 7. (11) 8. (3)
9. (1) 10. (4) 11. (2) 12. (48)

1. (292)

Given,

The tangent to the curve $x^2 + 2x - 4y + 9 = 0$ at the point $P(1, 3)$ on it meet the y -axis at A ,

So, tangent to circle at point $P(1, 3)$ is given by,

$$x + (x+1) - 2(y+3) + 9 = 0$$

$$\Rightarrow 2x - 2y = -4$$

So, on y -axis, $A(0, 2)$

And the line passing through P and parallel to the line $x - 3y = 6$ meet the parabola $y^2 = 4x$ at B ,

So, the equation of line through P will be $y - 3 = \frac{1}{3}(x - 1)$

$$\Rightarrow 3y = x + 8$$

Now finding intersection of $y^2 = 4x$ & $3y = x + 8$ we get,

$$B = (4, 4) \text{ \& } (16, 8),$$

Also B lies on the line $2x - 3y = 8$, so $(4, 4)$ will not satisfy,

Hence, $B = (16, 8)$

$$\text{So, } (AB)^2 = (0 - 16)^2 + (2 - 8)^2 = 256 + 36 = 292$$

2. (5)

Given,

A curve $y = f(x)$, $x \in (0, \infty)$ pass through the points $P\left(1, \frac{3}{2}\right)$ and $Q\left(a, \frac{1}{2}\right)$,

And the tangent at any point $R(b, f(b))$ to the given curve cuts the y -axis at the point $S(0, c)$ such that $bc = 3$,

Now tangent to the curve is given by,

$$Y - y = m(X - x), \quad m = \frac{dy}{dx}$$

Now at Y -axis, put $X = 0$ we get,

$$Y = y - mx \text{ or } c = y - mx \text{ as point } S(0, c) \text{ given,}$$

Now given $bc = 3$

$$\Rightarrow x(y - mx) = 3 \text{ \{here } b = x \text{ on the curve\}}$$

$$\Rightarrow y - \frac{xdy}{dx} = \frac{3}{x}$$

$$\Rightarrow \frac{ydx - xdy}{x^2} = \frac{3dx}{x} \cdot \frac{1}{x^2}$$

$$\Rightarrow d\left(\frac{y}{x}\right) = 3d\left(\frac{x^{-2}}{-2}\right)$$

Now integrating both side we get,

$$\Rightarrow \frac{y}{x} = \frac{3}{2x^2} + C,$$

Now given curve passes through $P\left(1, \frac{3}{2}\right)$, so $C = 0$

So, equation of curve will be $2xy = 3$

Now curve also passes through the point $Q\left(a, \frac{1}{2}\right)$

So, $a = 3$

Hence, $P\left(1, \frac{3}{2}\right)$, $Q\left(3, \frac{1}{2}\right)$

So, by distance formula we get, $(PQ)^2 = 4 + 1 = 5$

3. (5)

Let, $f(x) = x^5 - 20x^3 + 50x + 2$

Now differentiating the function $f(x)$ with respect to x we get,

$$f'(x) = 5x^4 - 60x^2 + 50$$

$$\Rightarrow f'(x) = 5(x^4 - 12x^2 + 10)$$

Now equating $f'(x) = 0$ we get,

$$x^4 - 12x^2 + 10 = 0$$

$$\Rightarrow x^2 = \frac{12 \pm \sqrt{144 - 40}}{2}$$

$$\Rightarrow x^2 = 6 \pm \sqrt{26}$$

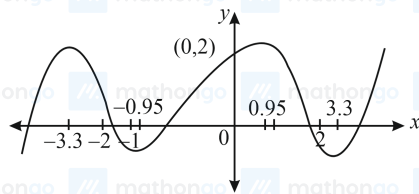
$$\Rightarrow x^2 \approx 11.1, 0.9$$

$$\Rightarrow x \approx \pm 3.31, \pm 0.95$$

Now finding the nature of function at integer value near by above value we get,

$$f(0) = 2, f(1) > 0, f(2) < 0, f(-1) < 0 \text{ \& } f(-2) > 0$$

Now plotting the graph by observing above values, we get,

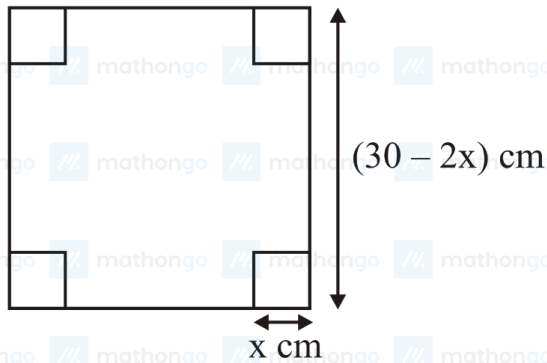


Hence, it will cross x -axis 5 times.

4. (1)

Given that, the side of square is 30 cm and x cm squares are cut off.

The required diagram is,



Now the dimensions of the cuboid formed will be

$$l(x) = 30 - 2x, b(x) = 30 - 2x \text{ and } h(x) = x.$$

The Volume of the cuboid will be $V(x) = (30 - 2x)^2(x)$

Now to get Maximum value,

$$\Rightarrow \frac{dV(x)}{dx} = 0$$

$$\Rightarrow 2(30 - 2x)(-2)x + (30 - 2x)^2(1) = 0$$

$$\Rightarrow (30 - 2x)(-4x + 30 - 2x) = 0$$

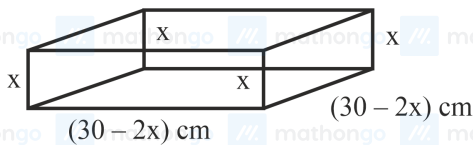
On simplifying we get,

$$\Rightarrow x = 15 \text{ cm, } 5 \text{ cm}$$

But x cannot be 15 cm as the volume becomes zero.

Hence $x = 5$ cm.

Now to find the surface area of the cuboid,



$$\text{Surface area will be } = (30 - 2x) \times x \times 4 + (30 - 2x)^2$$

$$= (30 - 2 \times 5) \times 5 \times 4 + (30 - 2 \times 5)^2$$

$$= 800 \text{ cm}^2.$$

Therefore, the required surface area will be 800 cm^2 .

5. (1)

Given,

$g(x) = f(x) + f(1-x)$ and $f''(x) > 0, x \in (0, 1)$. If g is decreasing in the interval $(0, \alpha)$ and increasing in the interval $(\alpha, 1)$,

Now solving,

$$g(x) = f(x) + f(1-x)$$

Now differentiating both side we get,

$$g'(x) = f'(x) - f'(1-x)$$

Differentiating again we get,

$$g''(x) = f''(x) + f''(1-x) > 0$$

So, $g'(x)$ is increasing as given $f''(x) > 0$

$$\Rightarrow g'(0) < g'(1)$$

$$\Rightarrow f'(0) - f'(1) < f'(1) - f'(0)$$

$$\Rightarrow f'(0) < f'(1)$$

Now finding $g'(x) = 0$ we get,

$$\Rightarrow f'(x) = f'(1-x)$$

$$\Rightarrow x = 1-x$$

$$\Rightarrow x = \frac{1}{2}$$

Now $g'(x)$ is positive for $x \in (0, \frac{1}{2})$

And $g'(x)$ is negative for $x \in (\frac{1}{2}, 1)$

$$\therefore \alpha = \frac{1}{2}$$

So,

$$\tan^{-1} 2\alpha + \tan^{-1} \left(\frac{1}{\alpha} \right) + \tan^{-1} \left(\frac{\alpha+1}{\alpha} \right)$$

$$= \tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3$$

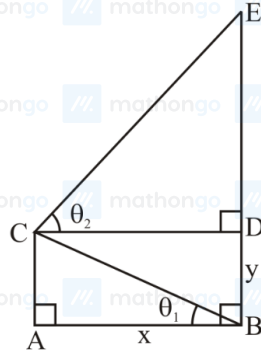
$$= \frac{\pi}{4} + \tan^{-1} \left(\frac{2+3}{1-6} \right)$$

$$= \frac{\pi}{4} + \tan^{-1}(-1)$$

$$= \frac{\pi}{4} + \frac{3\pi}{4} = \pi$$

6. (6)

Given,



$$\text{And } \theta_1 + \theta_2 = \frac{\pi}{2}$$

Now, Let $AB = x$, $BD = y$

Also given, $\sqrt{3}BE = 4AB$

$$\Rightarrow \sqrt{3}(y + DE) = 4x$$

$$\Rightarrow DE = \frac{4x}{\sqrt{3}} - y$$

Now given area of triangle $\Delta CAB = 2\sqrt{3} - 3$

$$\Rightarrow \frac{1}{2}xy = 2\sqrt{3} - 3$$

$$\Rightarrow y = \frac{4\sqrt{3}-6}{x}$$

Now finding,

$$\tan \theta_2 = \frac{\left(\frac{4x}{\sqrt{3}} - y\right)}{x} = \frac{4}{\sqrt{3}} - \frac{(4\sqrt{3}-6)}{x^2}$$

$$\tan \theta_1 = \frac{y}{x} = \frac{4\sqrt{3}-6}{x^2}$$

Now taking tan both of $\theta_1 + \theta_2 = \frac{\pi}{2}$ we get,

$$\tan \theta_1 \cdot \tan \theta_2 = 1$$

$$\Rightarrow \left(\frac{4}{\sqrt{3}} - \frac{4\sqrt{3}-6}{x^2}\right) \cdot \left(\frac{4\sqrt{3}-6}{x^2}\right) = 1$$

$$\Rightarrow \frac{4\sqrt{3}-6}{x^2} = \sqrt{3} \text{ or } \frac{1}{\sqrt{3}}$$

$$\text{So, } \frac{\theta_2}{\theta_1} \text{ is maximum when } \frac{4\sqrt{3}-6}{x^2} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow x^2 = \sqrt{3}(4\sqrt{3}-6)$$

$$\Rightarrow x^2 = 12 - 6\sqrt{3}$$

$$\Rightarrow x^2 = (3 - \sqrt{3})^2$$

$$\Rightarrow x = 3 - \sqrt{3} \text{ and } \theta_2 = 60^\circ$$

$$\Rightarrow \tan 60^\circ = \frac{DE}{CD} \Rightarrow DE = x \cdot \sqrt{3} = 3\sqrt{3} - 3$$

$$\text{And } \cos 60^\circ = \frac{CD}{CE} \Rightarrow CE = 2x = 6 - 2\sqrt{3}$$

So, the Perimeter of $\Delta CED = CD + DE + CE = 6$

7. (11)

Given,

The quadratic curve passing through the point $(-1, 0)$ and touching the line $y = x$ at $(1, 1)$ be $y = f(x)$,

Now let the curve be $y = ax^2 + bx + c$

Passes through $(-1, 0)$

$$\Rightarrow a - b + c = 0 \dots (i)$$

Passes through $(1, 1)$

$$\Rightarrow a + b + c = 1 \dots (ii)$$

Now given tangent to $y = x$ at $(1, 1)$,

$$\text{So, } \frac{dy}{dx} = 1$$

$$\Rightarrow 2a + b = 1 \dots (iii)$$

Solving all above three equations we get,

$$a = \frac{1}{4}, b = \frac{1}{2} \text{ and } c = \frac{1}{4}$$

Hence, the equation of curve is $y = \frac{x^2}{4} + \frac{x}{2} + \frac{1}{4}$

Now $(\alpha, \alpha + 1)$ lies on curve

$$\alpha + 1 = \frac{\alpha^2}{4} + \frac{\alpha}{2} + \frac{1}{4}$$

$$\Rightarrow (\alpha - 3)(\alpha + 1) = 0$$

$$\Rightarrow \alpha = 3$$

Point in first quadrant is $(3, 4)$

Now finding slope of tangent at $(3, 4)$ we get,

$$\left(\frac{dy}{dx}\right)_{(3,4)} = \left(\frac{x}{2} + \frac{1}{2}\right)_{(3,4)} = 2$$

$$\Rightarrow \text{Slope of normal} = -\frac{1}{2}$$

Hence, equation of normal will be, $y - 4 = -\frac{1}{2}(x - 3)$

So, x -intercept = 11

8. (3)

Given,

Domain and range of function,

$$f: [2, 4] \rightarrow \mathbb{R}$$

$$(x \log_e x) f'(x) + (\log_e x) f(x) + f(x) \geq 1, x \in [2, 4]$$

$$\Rightarrow (x \log_e x) \frac{d}{dx} f(x) + (\log_e x) f(x) \frac{d}{dx} (x) + x f(x) \frac{d}{dx} (\log_e x) \geq 1$$

$$\Rightarrow \frac{d}{dx} [x \ln x f(x)] \geq 1$$

$$\Rightarrow \frac{d}{dx} [x \ln x f(x)] \geq \frac{d}{dx} (x)$$

$$\Rightarrow \frac{d}{dx} [x \ln x f(x) - x] \geq 0$$

Hence, $h(x) = x(\ln x) f(x) - x$ is an increasing function,

$$\therefore h(x) \geq h(2), x \in [2, 4]$$

$$\Rightarrow x \ln x \times f(x) - x \geq 2 \ln 2 \times f(2) - 2$$

$$\Rightarrow x \ln x f(x) - x \geq \ln 2 - 2$$

Similarly, $h(x) \leq h(4)$

$$\Rightarrow x \ln x f(x) - x \leq \ln 4 - 4$$

$$\text{So, } \frac{\ln 2 - 2}{x \ln x} + \frac{1}{\ln x} \leq f(x) \leq \frac{\ln 4 - 4}{x \ln x} + \frac{1}{\ln x}$$

Now for $x \in [2, 4]$

$$\frac{\ln 4 - 4}{x \ln x} + \frac{1}{\ln x} \leq \frac{\ln 4 - 4}{2 \ln 2} + \frac{1}{\ln 2} = 1 - \frac{1}{\ln 2} < 1$$

$$\Rightarrow f(x) \leq 1 \text{ for } x \in [2, 4]$$

Now for $x \in [2, 4]$

$$\frac{\ln 2 - 2}{x \ln x} + \frac{1}{\ln x} \geq \frac{\ln 2 - 2}{4 \ln 4} + \frac{1}{\ln 4} = \frac{\ln 2 - 2}{4 \ln 4} + \frac{1}{\ln 4} = \frac{1}{8} + \frac{1}{4 \ln 2} > \frac{1}{8}$$

$$\text{Hence, } f(x) \geq \frac{1}{8}$$

Hence both A & B are correct.

Note this question was bonus in Jee Main 2023 April session, as LMVT on $f(x) \cdot x \ln x$ can't be satisfied.

Hence, no such $f(x)$ exist.

9. (1)

Given function is $f(x) = \left(\frac{\sqrt{3}e}{2\sin x}\right)^{\sin^2 x}$

For maxima or minima $f'(x) = 0$

$$f'(x) = f(x) \left[2\sin x \cos x \times \ln\left(\frac{\sqrt{3}e}{2\sin x}\right) + \sin^2 x \times \frac{1 \times 2 \sin x}{\sqrt{3}e} \times \frac{\sqrt{3}e}{2} \left(-\frac{1}{\sin^2 x} \times \cos x\right) \right]$$

$$\Rightarrow f(x) \left[\sin 2x \ln\left(\frac{\sqrt{3}e}{2\sin x}\right) - \sin x \cos x \right] = 0$$

Now on equation we get, $\sin 2x = 0$ (not possible)

$$\text{So, } \ln\left(\frac{\sqrt{3}e}{2\sin x}\right) = +\frac{1}{2}$$

$$\Rightarrow \frac{\sqrt{3} \times \sqrt{e}}{2\sin x} = e^{\frac{1}{2}}$$

$$\Rightarrow \sin x = \frac{\sqrt{3}}{2}$$

$$\Rightarrow f_{\max} = (e)^{\frac{3}{8}} = \frac{e^{\frac{11}{8}}}{e} \Rightarrow k = e^{\frac{11}{8}}$$

$$\Rightarrow \left(\frac{k}{e}\right)^8 + \frac{k^8}{e^5} + k^8 = e^3 + e^6 + e^{11}$$

10. (4)

$$\text{Let } f(x) = x - 2\sin x \cos x + \frac{1}{3}\sin 3x$$

$$\Rightarrow f'(x) = 1 - 2\cos 2x + \cos 3x$$

$$\Rightarrow f''(x) = 4\sin 2x - 3\sin 3x$$

For maxima/minima $f'(x) = 0$

$$\Rightarrow 1 - 2(2\cos^2 x - 1) + 4\cos^3 x - 3\cos x = 0$$

$$\Rightarrow (2\cos x + \sqrt{3})(2\cos x - \sqrt{3})(\cos x - 1) = 0$$

$$\cos x = \frac{-\sqrt{3}}{2}, \frac{\sqrt{3}}{2}, 1$$

$$x = \frac{5\pi}{6}, \frac{\pi}{6}, 0$$

$$f''\left(\frac{5\pi}{6}\right) = -2\sqrt{3} - \sqrt{3} < 0$$

$$f''\left(\frac{\pi}{6}\right) = 2\sqrt{3} - \sqrt{3} > 0$$

$$f''(0) = 0$$

So $x = \frac{5\pi}{6}$ is local maxima point

$$\text{Maximum value of } f(x) = f\left(\frac{5\pi}{6}\right) = \frac{5\pi}{6} + \frac{\sqrt{3}}{2} + \frac{1}{3}$$

$$= \frac{5\pi + 2 + 3\sqrt{3}}{6}$$

Hence this is the correct option.

11. (2)

Given,

$$x|x-1| + |x+2| + a = 0$$

Now taking, Case I: $x < -2$, we get

$$-x^2 + x - x - 2 + a = 0$$

$$\Rightarrow a = x^2 + 2$$

And $y = x^2 + 2$ is decreasing $\forall x \in (-\infty, -2)$

Now taking Case II: $-2 \leq x < 1$ we get,

$$-x^2 + x + x + 2 + a = 0$$

$$\Rightarrow a = x^2 - 2x - 2$$

$$\text{And } y = x^2 - 2x - 2$$

$$\text{So, } \frac{dy}{dx} = 2x - 1 \leq 0 \forall x \in [-2, 1)$$

Hence, y is decreasing $\forall x \in [-2, 1)$

Now taking Case III: $x \geq 1$ we get,

$$x^2 - x + x + 2 + a = 0$$

$$\Rightarrow a = -(x^2 + 2)$$

And $y = -(x^2 + 2)$ is decreasing $\forall x \in [1, \infty)$

Hence, from all the cases we can say that nature of function is continuously decreasing so, it will cut x -axis only one time,

$\therefore$ Exactly one real root $\forall a \in \mathbb{R}$

12. (48)

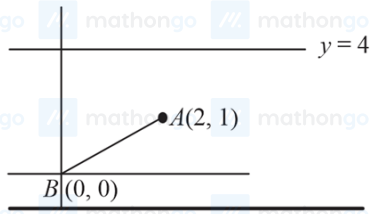
Given,

The triangles with vertices $A(2, 1)$, $B(0, 0)$ and $C(t, 4)$, $t \in [0, 4]$,

And the maximum and the minimum perimeters of such triangles are obtained at $t = \alpha$ and $t = \beta$ respectively,

Now to minimise $CA + CB$, in below diagram take image of B in $y = 4$,

We get, $B' = (0, 8)$



Now finding, equation of AB' we get,

$$y - 8 = \frac{-7}{2}(x - 0)$$

Now, putting $y = 4$ in above equation we get,

$$-4 = \frac{-7}{2}(x)$$

$$\Rightarrow x = \frac{8}{7} \Rightarrow \beta = \frac{8}{7}$$

Now, maximum perimeter will be possible if $\alpha = 0$ or 4

Now taking $\alpha = 0$ we get,

$$AB = \sqrt{5}, BC = 4 \text{ \& } AC = \sqrt{13}$$

Now when $\alpha = 4$ we get,

$$AB = \sqrt{5}, BC = 4\sqrt{2} \text{ \& } AC = \sqrt{13}$$

Now on comparing the perimeter we get, maximum perimeter at $\alpha = 4$

$$\text{Hence, } 6\alpha + 21\beta = 48$$